

CS 354 Fall 2025

Lab 7: Running A Remote File Server And Remote File Access [20 pts]

Due: 11/08/2025 (Sat), 11:59 PM

You Will Learn

How an operating system provides read/write access to remote files.

How to run a server and configure Xinu to access it.

Background

Remote file access involves two computers that communicate over a network. One computer stores files and runs a file server that allows the other computer to access the files. A second computer runs file access software that sends requests to the server to read and write files.

In the Xinu remote file system, the server runs on a Linux machine and the file access software runs in Xinu. The access software on Xinu must be configured so it knows the computer on which the server is running (i.e., the Xinu system must be told where the server is running because it does not have a way to find the location).

Multiple servers can run on a given Linux computer at the same time. To distinguish among them, each is assigned a unique integer ID known as a protocol port number (port number). The Xinu access software must also be configured with the port number that the server is using.

Review the Xinu NAMESPACE and how it maps files.

Source code

ssh into the front-end computer you have chosen, and to avoid running out of space, remove an old copy of Xinu.

Start from a fresh copy of `CS354_lab7.tar.gz` by running the command:

```
tar zxvf /homes/cs354/CS354_lab7.tar.gz
```

- which will create a `lab7` directory that contains three subdirectories:
 - `rfserver` - source and object files for a Xinu remote file server
 - `myfiles` - a directory where you will run the server
 - `xinu-bbb` - a copy of xinu you will use for the lab

The subdirectories do not really need to be under a single directory -- we just decided to put them in a single tarball.

Summary Of What You Will Do

Obtain a protocol port number for you to use (so you don't interfere with other students). Use your port number to run a remote file server on a Xinu front-end computer.

Configure Xinu remote access software to contact your remote file server (i.e., specify the computer on which the server is running and port number the server is using).

Boot Xinu and use the Xinu shell to verify that it can read and write files on the remote server.

Steps You Will Take

1. Listen to the TAs give hints about
 - Setting up the file server and
 - Configuring the Server information in Xinu
2. Retrieve the protocol port number you were assigned via email.
3. Choose a front-end computer in the Xinu lab where you will run your server and remember the name (e.g., `xinu $\mathbf{N}$ .cs.purdue.edu`, $\mathbf{N}$ could be `01, 02, \dots, 21`).
4. Enter the `lab7` folder (untared previously)
5. Configure the remote file server to use your assigned port number as the default

- Change to subdirectory `rfserver` and edit file `rfilesys.h`
- Replace the define for `RF_SERVER_PORT` with your assigned port number
- Type `make clean; make`
 - to compile a binary named `rfserver`

6. Start the remote file server running in the `myfiles` directory.

- Change to the directory `lab7/myfiles`
- Enter the following command


```
../rfserver/rfserver &
```

7. Configure Xinu to use your remote file server

- Change to directory `lab7/xinu-bbb/compile`
- Edit `../include/rfilesys.h` and change `RF_SERVER` from `"example.com"` to the front-end you are using (`"xinuN.cs.purdue.edu"`, remember to replace `N` with actual number) and change `RF_SERVER_PORT` to your assigned port number
- Edit `main.c` and insert the following before the shell invocation


```
int32 x = open(NAMESPACE, "/remote/Bird", "r");
int ch = getc(x);
while (ch != EOF) {
    putc(CONSOLE, ch);
    ch = getc(x);
}
close(x);
return 0;
```
- Run `make clean; make` to rebuild Xinu

8. Run the version of Xinu and answer question 1.

9. Remove the `return 0;` statement from `main` and rerun Xinu (which will cause Xinu to execute the shell -When the shell prompt appears, type

```
ls ~
```

and answer question 2.

10. At the shell prompt, enter

```
echo hello there #1 > /remote/Q
echo hello there #2 > Q
```

`cat /remote/Q`
and answer question 3.

11. At the shell prompt, enter

`cat Q`
and answer questions 4 and 5.

12. Change `cs-console` to command mode and quit (`q` command). Then use `cs-console` to rerun your Xinu. At the shell prompt, enter

`cat /remote/Q`
and answer question 6.

13. At the shell prompt, enter

`cat Q`
and answer questions 7.

14. Use `cs-console` command `z` to suspend `cs-console`. While `cs-console` is suspended, kill your `rfserver` and then restart it. bring `cs-console` into foreground and at the shell prompt, enter

`cat /remote/Q`
and answer question 8.

15. At the shell prompt, enter

`date | tee /remote/TIMESTAMP`
and answer question 9.

16. Note the date printed in step 15, suspend `cs-console`, list the files in `myfiles`, look at the contents of file `TIMESTAMP`, if it exists, and answer question 10.

Questions

Place **short answers** to the following questions in a file named `questions.pdf`. In the case of using document software (e.g., Word), please make sure to convert the file to **PDF**.

1. What is the output in step 8?
2. What is the output in step 9?
3. What is the output in step 10?

4. What is the output in step 11?
5. Explain the results in the previous two questions
6. What is the output in step 12?
7. What is the output in step 13?
8. What is the output in step 14?
9. What is the output in step 15?
10. What files are present in `myfiles`? If a `TIMESTAMP` file exists, is the date identical to the date printed in step 15? Explain why or why not.

Submission

General instructions:

When implementing code in the labs, please maintain separate versions/copies of code so that mistakes such as unintentional overwriting or deletion of code is prevented. This is in addition to the efficiency that such organization provides. You may use any number of version control systems such as GIT and RCS. Please make sure that your code is protected from public access. For example, when using GIT, use `git` that manages code locally instead of its on-line counterpart `github`. If you prefer not to use version control tools, you may just use manual copy to keep track of different versions required for development and testing. More vigilance and discipline may be required when doing so.

You must submit a directory named `lab7` that contains both required files:

- Your final version of `main.c` from Xinu
- `questions.pdf`, containing your answers

Go to the directory where `lab7` is a subdirectory.

For example, if `/homes/alice/cs354/lab7` is your directory structure, go to `/homes/alice/cs354`.

Type the following command to submit the directory with `turnin`:

```
turnin -c cs354 -p lab7 lab7
```

Be sure the files inside the directory are named exactly `main.c` and `questions.pdf`.

You can check/list the submitted files using

```
turnin -c cs354 -p lab7 -v
```